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Test

IPCV

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Ans1 Frequency domain filtering involves transforming an image into its frequency components using the Fourier Transform

→ Low-Pass Filtering → high frequency noise and edges smoothens the image (reduce sharpness)

→ High pass filtering - Low frequency blur like smooth areas sharpen the edges.

Ans 2 → Step by Step process to perform filtering.

1. Convert the input image to the frequency domain using using Fast
2. Shift the FFT output to center the low frequencies
3. Multiply the frequency spectrum by desired filtering
4. Perfor Inverse FFT shift to rearrange spectrum
5. Apply the inverse FFT to obtain the filtered spatial domain image.

Ans 3 → Low-pass-filtering → high frequencies noise and edges smoothens the image (reduce) sharpness.

High pass → Low frequency blur like smooth areas sharpen the edges & can amplify noise

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Q 4 → Image degradation in blur and noise.

→ Noise → The unwanted disturbances or spots on an image is called noise.
Poisson noise, Salt - Pepper noise,

→ Blur - convolution with point - spread function which removes high frequencies reduce image.

Image Degradation formula

$$g = f * h + n \text{ — noise}$$

blur.